

Lookup

Functions

1. VLOOKUP()

*What is VLOOKUP?

VLOOKUP() searches for a value in the **first column of a table** and returns a corresponding value from another column.

*Remember:

V = Vertical

It searches **top → bottom**.

*Syntax

=VLOOKUP(lookup_value,table_array,col_index_num,[range_lookup])

*Parameters

Argument	Meaning
lookup_value	Value you want to find
table_array	Table containing the data
col_index_num	Column number to return
range_lookup	Exact or approximate match

2. HLOOKUP()

*What is HLOOKUP?

HLOOKUP() searches **horizontally**.

*Remember:

H = Horizontal

It searches:

left → right

Example Dataset

Imagine your data is arranged like this:

	B	C	D	E
Employee	Rahul	Priya	Amit	Neha
Salary	45000	38000	52000	60000
Experience	2	3	4	5
Department	IT	HR	IT	IT

Suppose H2 contains:

Amit

You want his salary.

*Syntax

=HLOOKUP(lookup_value,table_array,row_index_num,[range_lookup])

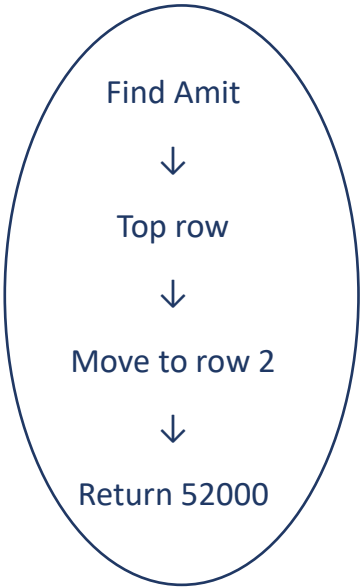
Formula:

=HLOOKUP(H2,B1:E4,2,FALSE)

*Result

52000

*How it works



*HLOOKUP Example — Experience

=HLOOKUP(H2,B1:E4,3,FALSE)

Result:

4

HLOOKUP vs VLOOKUP

VLOOKUP	HLOOKUP
Vertical	Horizontal
Searches first column	Searches first row
Returns column value	Returns row value
More commonly used	Less commonly used

3. XLOOKUP()

★ This is extremely important.

XLOOKUP() is the modern lookup function and is generally more flexible than VLOOKUP.

It can:

- Look left
- Look right
- Return exact matches
- Handle missing values
- Search vertically or horizontally
- Avoid column-number counting

*Syntax

=XLOOKUP(lookup_value,lookup_array,return_array,[if_not_found])

*Example — Find Salary

Suppose:

H2 = E003

Employee ID:

A2:A11

Salary:

E2:E11

Formula:

=XLOOKUP(H2,A2:A11,E2:E11)

*Result

52000

XLOOKUP — Find Name

=XLOOKUP(H2,A2:A11,B2:B11)

Result:

Amit

XLOOKUP — Find Department

=XLOOKUP(H2,A2:A11,C2:C11)

Result:

IT

★ XLOOKUP with "Not Found"

Suppose H2 contains:

E999

There is no such employee.

Instead of:

#N/A

you can write:

=XLOOKUP(H2,A2:A11,E2:E11,"Employee Not Found")

Result:

Employee Not Found

XLOOKUP Can Look LEFT

Suppose:

Name	Employee_ID	Department
Rahul	E001	IT
Priya	E002	HR
Amit	E003	IT

You want:

Employee ID for Amit

With XLOOKUP:

=XLOOKUP(H2,A2:A4,B2:B4)

if H2 contains Amit.

It returns:

E003

This is something standard VLOOKUP cannot easily do.

4. INDEX()

Now we move to a different way of looking up data.

What does INDEX do?

INDEX() returns a value from a specific position in a range.

Think:

"Give me the value at this row/column position."

*Syntax

For a one-dimensional range:

=INDEX(array,row_num)

For a two-dimensional range:

=INDEX(array,row_num,column_num)

*Example

Suppose:

B2:B11 = Employee Names

Formula:

=INDEX(B2:B11,3)

Excel returns the **3rd value** from the range.

Result

Amit

Because:

1 → Rahul

2 → Priya

3 → Amit

INDEX with Two Dimensions

Suppose:

B2:E11

contains:

Name	Department	City	Salary
Rahul	IT	Pune	45000
Priya	HR	Mumbai	38000
Amit	IT	Nashik	52000

Formula:

=INDEX(B2:E11,3,4)

Meaning:

3rd row

4th column

Result:

52000

5. MATCH()

***What does MATCH do?**

MATCH() finds the **position** of a value in a range.

Important:

MATCH doesn't normally return the value. It returns its position.

***Syntax**

=MATCH(lookup_value,lookup_array,[match_type])

For exact match:

=MATCH(H2,A2:A11,0)

Suppose:

H2 = E003

Employee IDs:

E001

E002

E003

E004

...

Result

3

Because E003 is the 3rd item in the range.

MATCH Example

Find the position of:

Neha

in B2:B11:

=MATCH("Neha",B2:B11,0)

Result:

6

INDEX vs MATCH

This is extremely important.

INDEX

"Give me the value at position X."

MATCH

"Tell me the position of X."

Example:

=MATCH("Neha",B2:B11,0)

returns:

6

Then:

=INDEX(E2:E11,6)

returns:

60000

6. INDEX + MATCH

This is a classic Data Analyst interview topic.

It combines:

MATCH → finds position

INDEX → returns value

Business Question

Find the salary of employee E003.

Step 1 — MATCH

=MATCH(H2,A2:A11,0)

If H2 = E003

Result:

3

Step 2 — INDEX

=INDEX(E2:E11,3)

Result:

52000

★ Combined Formula

Instead of two separate formulas:

=INDEX(E2:E11,MATCH(H2,A2:A11,0))

How Excel processes it:

H2 = E003

↓

MATCH(H2,A2:A11,0)

↓

3

↓

INDEX(E2:E11,3)

↓

52000

INDEX + MATCH — Find Department

=INDEX(C2:C11,MATCH(H2,A2:A11,0))

If H2 = E003:

IT

INDEX + MATCH — Find Name

=INDEX(B2:B11,MATCH(H2,A2:A11,0))

Result:

Amit

INDEX + MATCH — Find Experience

=INDEX(F2:F11,MATCH(H2,A2:A11,0))

Result:

4

REAL DATASET — SALES ORDERS

Sheet 1: Orders

Order_ID	Customer_ID	Product_ID	Quantity
O001	C001	P001	2
O002	C003	P002	5
O003	C002	P003	1
O004	C001	P002	3
O005	C004	P004	2
O006	C005	P001	1

Sheet 2: Customers

Customer_ID	Customer_Name	City	Segment
C001	Rahul	Pune	Premium
C002	Priya	Mumbai	Standard
C003	Amit	Nashik	Premium
C004	Neha	Pune	Standard
C005	Karan	Mumbai	Premium

Sheet 3: Products

Product_ID	Product_Name	Category	Price
P001	Laptop	Electronics	60000
P002	Mouse	Electronics	4000
P003	Chair	Furniture	14000
P004	Desk	Furniture	24000

BUSINESS PROBLEM

The Orders table only has IDs.

Management wants:

Order ID

Customer Name

Customer City

Customer Segment

Product Name

Category

Price

This is exactly where lookup functions become useful.

Using VLOOKUP

In Orders sheet, suppose:

B2 = C001

Find customer name:

=VLOOKUP(B2,Customers!A2:D6,2,FALSE)

Result:

Rahul

Customer city:

=VLOOKUP(B2,Customers!A2:D6,3,FALSE)

Result:

Pune

Customer segment:

=VLOOKUP(B2,Customers!A2:D6,4,FALSE)

Result:

Premium

Using XLOOKUP

Customer name:

=XLOOKUP(B2,Customers!A2:A6,Customers!B2:B6,"Not Found")

Customer city:

=XLOOKUP(B2,Customers!A2:A6,Customers!C2:C6,"Not Found")

Customer segment:

=XLOOKUP(B2,Customers!A2:A6,Customers!D2:D6,"Not Found")

Product Lookup

Suppose C2 contains:

P001

Find product name:

=XLOOKUP(C2,Products!A2:A5,Products!B2:B5,"Product Not Found")

Result:

Laptop

Find price:

=XLOOKUP(C2,Products!A2:A5,Products!D2:D5,"Product Not Found")

Result:

₹60,000

Calculate Order Revenue

Once you've retrieved the product price:

Revenue = Quantity × Price

If:

D2 = Quantity

H2 = Price

then:

=D2*H2

For O001:

2 × ₹60,000

Revenue

₹1,20,000

This is how lookup functions become part of actual **business analysis**.

CASE STUDY

Question

You have two tables.

Employee Master

Employee_ID	Name	Department	Salary
E001	Rahul	IT	45000
E002	Priya	HR	38000
E003	Amit	IT	52000

Bonus Table

Employee_ID	Bonus
E001	5000
E002	3000
E003	7000

HR gives you Employee ID E003.

Question:

How would you retrieve Amit's bonus?

XLOOKUP

=XLOOKUP(H2,Bonus!A2:A4,Bonus!B2:B4,"Not Found")

Result:

₹7,000

VLOOKUP

=VLOOKUP(H2,Bonus!A2:B4,2,FALSE)

EMPLOYEE HR LOOKUP SYSTEM

Create:

Employee_HR_Lookup.xlsx

Sheet 1 — Employee_Master

Your 10 employee dataset.

Sheet 2 — Employee_Search

Create this interface:

Field	Value
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Employee ID	E003
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Employee Name

Department

City

Salary

Experience

Use XLOOKUP.

Employee Name

=XLOOKUP(B2,Employee_Master!A2:A11,Employee_Master!B2:B11,"Not Found")

Department

=XLOOKUP(B2,Employee_Master!A2:A11,Employee_Master!C2:C11,"Not Found")

City

=XLOOKUP(B2,Employee_Master!A2:A11,Employee_Master!D2:D11,"Not Found")

Salary

=XLOOKUP(B2,Employee_Master!A2:A11,Employee_Master!E2:E11,"Not Found")

Experience

=XLOOKUP(B2,Employee_Master!A2:A11,Employee_Master!F2:F11,"Not Found")

Now you have a mini **HR employee search tool**.

SALES ORDER ENRICHMENT

Use the Orders + Customers + Products datasets.

Create:

Order_ID	Customer_ID	Customer_Name	City	Segment	Product_ID	Product_Name	Category	Price	Quantity	Revenue
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Use XLOOKUP() to populate:

- Customer Name
- City
- Segment
- Product Name
- Category
- Price

Then:

=Quantity*Price
for Revenue.